

M003 Extension — English Worksheet (Beginner)

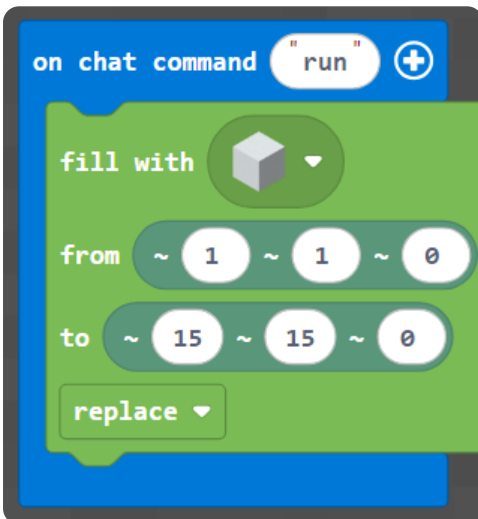
Topic: Pixel-Art Mobs (x, y) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after Week 3) ·

Time: about 30 minutes

This is a **bonus picture challenge** between Week 3 and Week 4. You copy a mob onto your wall using (x, y). x is how far **across**, y is how far **up**. Read the picture and place each block at its (x, y).

First, build your canvas. Run this to make a blank 15 × 15 wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too and the blocks land in the wrong place. Same spot every time = a clean picture.

1 · Predict 🧠

Read the steps. Before you imagine it happening, circle or write your answer.

```
place pink block at (4, 13)
place pink block at (5, 13)
place pink block at (6, 13)
```

Do these blocks make a line going across? Circle one: across · up

Which number stays the same — x or y?


```
place black block at (8, 7)
place black block at (8, 8)
```

These share $x = 8$ and y grows. Do they go across or up? Circle one: across · up

2 · Find the Difference

Each pair shows the **correct** line first, then a **broken** one. Circle what's different and write one short sentence about the bug.

Pair A — The axolotl's two eyes should be at **different** spots: (6, 10) and (11, 10).

```
# correct
place black block at (6, 10)
place black block at (11, 10)
```

```
# broken
place black block at (6, 10)
place black block at (6, 10)
```

What is wrong? Where do both blocks land?

Pair B — The dragon's eye should be at (8, 10).

```
# correct
place purple block at (8, 10)
```

```
# broken
place purple block at (10, 8)
```

What is wrong? Which two numbers got swapped?

3 · Fill the Gap

To move **up** a column, the **second** number (y) has to grow. Fill in the missing numbers so this line goes up at $x = 8$.

```
place black block at (8, 7)
place black block at (8, __)
place black block at (8, __)
```

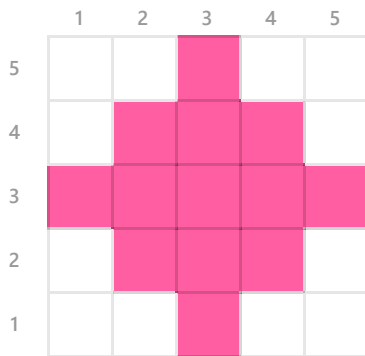
Word bank for the blanks: 8 · 9

Write the two missing numbers:

home spot (the red block).

Warm up first. Copy this small axolotl frill onto your wall. Read each pink square's (x, y) and place a pink block there.

 Axolotl frill



Now build the big  axolotl from the main worksheet grid. Read each colored square and place that color at its (x, y).

When your picture is finished, send a photo of it, then explain what you did. Use these sentence starters — write 3 to 5 sentences total.

First, I built the ...

The lowest block in my picture is at $(x = \dots, y = \dots)$.

To go across I changed the ... number; to go up I changed the ... number.

One time a block landed in the wrong place because ...

5 · Record Your Walkthrough 🎥

Take a short video on your phone (or a parent's phone) while you show your axolotl on the wall. Try to use these words: **x, y, across, up, home spot**.

1. Show your home spot (the red block).
2. Show your finished axolotl.
3. Point at one block and read its coordinate: "this one is at x ..., y ...".

Write what you will say in your video.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.